

Vinay Pepakayala

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Software Engineer — Java, Scalable Services & Distributed Systems

SUMMARY

Software Engineer candidate with hands-on experience in Java, REST APIs, Kubernetes automation, distributed-system diagnostics, and data-driven applications. Strong foundation in OOP, SOLID principles, design patterns, data structures, algorithms, DBMS, operating systems, computer networks, and system design; 1800+ LeetCode rating and GATE DA AIR 1173.

EDUCATION

National Institute of Technology Raipur, Chhattisgarh

2022–2026

B.Tech, Electronics & Communication Engineering | CGPA: 8.2

Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, System Design, Computer Architecture

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, SQL, Bash

Backend: Java, Spring Boot, RESTful APIs, Node.js, Express.js, OOP, SOLID, design patterns

Data & Testing: MySQL, MongoDB, API reliability

Platforms: Docker, Kubernetes, Helm, Prometheus, Alertmanager, Linux, Git, Maven, CI/CD

AI/ML: PyTorch, CNNs, OpenCV, model evaluation, data preprocessing, Azure AI Essentials, GenAI tools

EXPERIENCE

SDE Intern, Oracle

Jan 2026–Present

Diagnostics Collection Utility — Kubernetes supportability automation for Oracle ECE

- Contributed to the Java-based Oracle ECE charging repository, working across Maven modules, JVM diagnostics, Kubernetes deployment artifacts, and supportability workflows.
- Built alert-triggered diagnostics automation to collect pod logs, JVM data, thread dumps, heap/profiling artifacts, configuration files, and runtime evidence for faster production triage.
- Integrated Prometheus Alertmanager with Python orchestration and Bash execution while aligning collection workflows with Java service runtime behavior and operational reliability needs.
- Configured Kubernetes Services, RBAC, Helm values/templates, and monitoring-agent wiring for secure in-pod execution and reusable artifact collection across ECE deployments.
- Debugged porting issues by tracing failures across Java service assumptions, scripts, deployment config, and runtime dependencies, then validating fixes through end-to-end container tests.

Research Intern, NIT Warangal — Prof. Ch. V. Rama Rao

May–Jul 2024

- Developed a CNN-based steering-angle prediction pipeline on 2K+ Comma.ai samples, achieving MSE 0.29 and R² 0.87 for autonomous-driving perception research.
- Improved model reliability with data preprocessing, augmentation, Spatial Pyramid Pooling, validation metrics, and iterative testing across varied road scenes.
- Deployed optimized inference on NVIDIA Jetson Nano with 182 ms latency, gaining experience in model deployment, runtime constraints, and performance monitoring.

PROJECTS

MediTracker | MERN, Node.js, Express.js, React.js, MongoDB, ESP32

Live | GitHub

- Designed a medication-adherence platform connecting an ESP32 smart pillbox with REST APIs, MongoDB storage, and a React caregiver dashboard.
- Implemented reminder scheduling, intake acknowledgments, missed-dose tracking, sensor-state synchronization, and analytics for 100+ daily medication records.
- Improved medication visibility to 96% through real-time logs, adherence history, and caregiver-facing dashboard views.

Face Recognition Attendance System | OpenCV, Streamlit, Python

GitHub

- Designed a modular attendance system with separate registration, recognition, storage, and reporting flows for maintainable records.
- Implemented OpenCV preprocessing and recognition logic with database-backed attendance storage for reliable identity matching.
- Reduced manual attendance effort by 80% through automated verification, timestamped records, and result review workflows.

ACHIEVEMENTS & LEADERSHIP

- LeetCode:** 1800+ rating; practiced DSA patterns across arrays, graphs, DP, trees, heaps, and greedy algorithms.
- GATE 2025 Data Science & AI:** AIR 1173, Score 557.
- ZS Campus Beats Tech Challenge 2025:** 1st Runner-Up among 500+ teams.
- Research Publication:** “Vision-Based Steering Angle and Speed Prediction for Autonomous Vehicles,” ITI 2025. Link
- Led the ESC Design Domain; Microsoft Azure AI Essentials — LinkedIn Learning.